EXP 12: Implement Iris Flower Classification using KNN

CODING:

```
import math

# Training Dataset
X = [
   [5.1, 3.5, 1.4, 0.2],
   [4.9, 3.0, 1.4, 0.2],
   [5.0, 3.4, 1.5, 0.2],
   [6.4, 3.2, 4.5, 1.5],
   [6.0, 2.9, 4.5, 1.5],
   [5.9, 3.0, 4.2, 1.5],
   [6.5, 3.0, 5.5, 1.8],
   [6.7, 3.1, 5.6, 2.4],
   [6.3, 2.9, 5.6, 1.8]
]

Y = [
   'Setosa', 'Setosa', 'Setosa',
   'Versicolor', 'Versicolor', 'Versicolor',
   'Virginica', 'Virginica', 'Virginica'
]

# Calculate Euclidean Distance
def distance(a, b):
   total = 0
```

```
    for i in range(len(a)):
        total += (a[i] - b[i]) ** 2

    return math.sqrt(total)

# KNN Classification
def knn(sample, k=3):
    distances = []

    for i in range(len(X)):
        d = distance(sample, X[i])
        distances.append((d, Y[i]))

    distances.sort(key=lambda x: x[0])

    neighbours = []

    for i in range(k):
        neighbours.append(distances[i][1])

    prediction = max(set(neighbours),
                     key=neighbours.count)

    return prediction

# Display Training Dataset
print("Iris Flower Classification")
print("\nTraining Dataset:")
```

```
for i in range(len(X)):
    print(X[i], ":", Y[i])

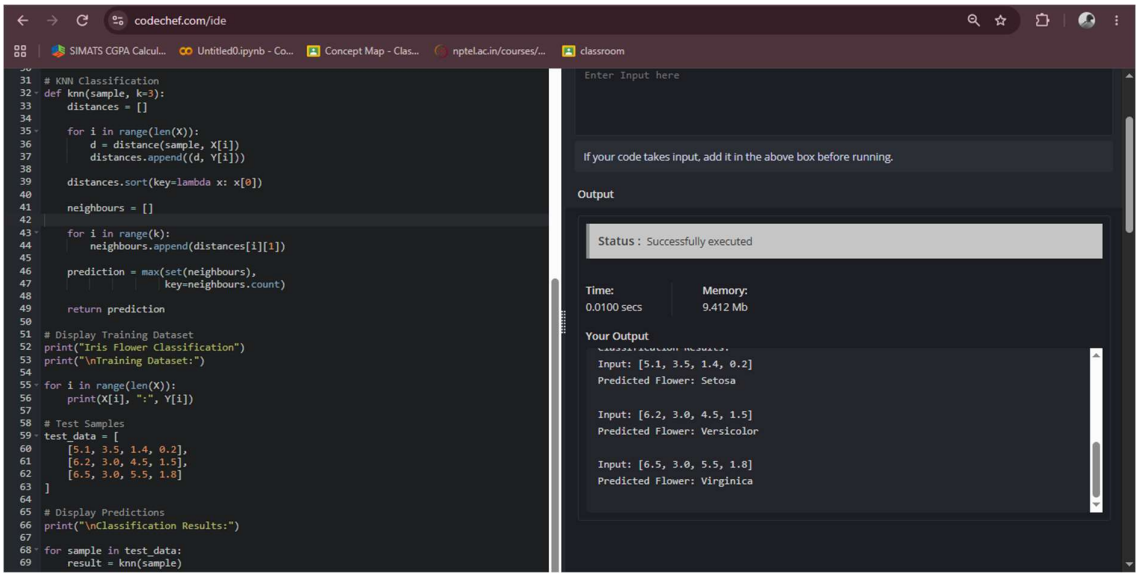
# Test Samples
test_data = [
    [5.1, 3.5, 1.4, 0.2],
    [6.2, 3.0, 4.5, 1.5],
    [6.5, 3.0, 5.5, 1.8]
]

# Display Predictions
print("\nClassification Results:")

for sample in test_data:
    result = knn(sample)

    print("Input:", sample)
    print("Predicted Flower:", result)
    print()
```

OUTPUT:


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KNN Classification
def knn(sample, k=3):
distances = []
for i in range(len(X)):
d = distance(sample, X[i])
distances.append((d, Y[i]))
distances.sort(key=lambda x: x[0])
neighbours = []
for i in range(k):
neighbours.append(distances[i][1])
prediction = max(set(neighbours),
key=neighbours.count)
return prediction
Display Training Dataset
print("Iris Flower Classification")
print("\nTraining Dataset:")
for i in range(len(X)):
print(X[i], ":", Y[i])
Test Samples
test_data = [
[5.1, 3.5, 1.4, 0.2],
[6.2, 3.0, 4.5, 1.5],
[6.5, 3.0, 5.5, 1.8]
]
Display Predictions
print("\nClassification Results:")
for sample in test_data:
result = knn(sample)
Enter Input here
If your code takes input, add it in the above box before running.
Output
Status : Successfully executed
Time:
0.0100 secs
Memory:
9.412 Mb
Your Output
Input: [5.1, 3.5, 1.4, 0.2]
Predicted Flower: Setosa
Input: [6.2, 3.0, 4.5, 1.5]
Predicted Flower: Versicolor
Input: [6.5, 3.0, 5.5, 1.8]
Predicted Flower: Virginica